ShrubMap Data Challenge

End-to-End Shrub Segmentation Pipeline

High-Resolution Shrub Mapping and Machine Learning Pipeline for Ecosystem Characterization

Toward a Public Health Perspective: From Shrub Cover to Wildfire Smoke and PM2.5 Exposure

Wildfire Science & Technology Commons, University of California, San Diego

Sami Bahig, AI Engineer & ML Researcher | MD, MSc

Submission: April 19, 2026

Best Performance

Best Model	IoU	F1	Recall	Precision
ResNet34-UNet 192×192 ★	**0.8397**	0.9055	0.9585	0.8579
Ensemble 2×ResNet34-UNet 192×192	0.8320	0.9083	0.9607	0.8613

Question 1 — Complete Pipeline, Data & Preprocessing

> ***Executive Summary —*** *Our pipeline integrates 0.6m NAIP multispectral imagery with 299 TLS LiDAR-derived shrub masks as ground truth labels across 6 ecologically diverse California sites. TLS LiDAR is used exclusively as a labeling source, polygon masks are reprojected to the NAIP grid (EPSG:26910) and merged by union into a per-site binary label map. Feature engineering produces 12 channels per pixel: 4 raw NAIP bands (R, G, B, NIR), 6 spectral indices (NDVI, EVI, TGI, NDWI, Brightness, VARI), and 2 texture features (local NDVI variance 5×5 window, NIR Shannon entropy). Patches are extracted at 64×64 (stride=16, min_shrub_ratio=1%), yielding 6,566 base patches across 6 sites for classical ML baselines. For deep learning, 192×192 patches were extracted directly (stride=32, min_shrub_ratio=0.3%), yielding 668 base patches across 5 sites (dl-bliss excluded due to missing NAIP), then normalized per-channel via percentile clipping (p1–p99). A full augmentation pipeline (×8, MixUp, CutMix, Copy-Paste Shrub, spectral jitter, channel dropout) was designed and validated to expand the dataset to ~31,000 patches; due to JupyterHub server instability, the final training used a simpler ×9 geometric augmentation on 64×64 patches yielding 6,680 samples. The full pipeline is implemented and documented in 02_preprocessing.ipynb. Class imbalance (1:21) is addressed via SMOTE for ML models and Dice+BCE loss with pos_weight=21 for DL models.*

Our pipeline integrates 0.6m NAIP multispectral imagery with 299 TLS LiDAR-derived shrub masks as ground truth labels across 6 ecologically diverse California sites (Calaveras Big Trees, DL Bliss, Independence Lake, Pacific Union College, Sedgwick, Shaver Lake), spanning Mediterranean chaparral, oak savanna, riparian forest, and subalpine biomes. TLS LiDAR is used exclusively as a precision labeling source, polygon masks are reprojected to the NAIP grid (EPSG:26910) via rasterio and merged by union into a per-site binary label map, ensuring sub-meter alignment between imagery and ground truth.

Feature engineering produces 12 channels per pixel following the multi-modal design validated for vegetation segmentation by Kattenborn et al. (2021): 4 raw NAIP bands (R, G, B, NIR), 6 spectral indices (NDVI for vegetation vigor, EVI for soil-noise reduction, TGI for triangular greenness, NDWI for water exclusion, Brightness, VARI), and 2 texture features (local NDVI variance on a 5×5 window capturing surface roughness, NIR Shannon entropy on a disk of radius 3 capturing structural complexity). SHAP analysis confirms that texture_ent and NDWI are the most discriminative features for shrub detection. All channels are normalized per-channel via percentile clipping (p1–p99 → [0,1]) to handle extreme reflectance ranges across ecologically diverse sites.

For classical ML baselines, patch extraction uses a 64×64 base resolution (stride=16, min_shrub_ratio=1%), yielding 6,566 annotated patches across 6 sites (pixel-level flattening produces ~30M samples). For deep learning, 192×192 patches were extracted directly (stride=32, min_shrub_ratio=0.3%), yielding 668 base patches across 5 sites (dl-bliss excluded due to missing NAIP during this pipeline run), then upsampled and augmented ×9 to produce 6,680 training samples. The 192×192 resolution is shown in our experiments to improve IoU by +8.3 points over 128×128 patches, consistent with the spatial context requirements identified by Kattenborn et al. (2019) for shrub species segmentation from aerial imagery. Class imbalance (1:21 background-to-shrub ratio) is addressed via SMOTE for classical ML models (He et al., 2009) and combined Dice+BCE loss with pos_weight=21 for deep learning models (Lin et al., 2017), a combination that directly optimizes the F1-equivalent overlap metric while heavily penalizing false negatives. A full augmentation pipeline (×8, HorizontalFlip, VerticalFlip, RandomRotate90, ShiftScaleRotate, ElasticTransform, GridDistortion, CoarseDropout, GaussNoise, GaussianBlur, MixUp, CutMix, Copy-Paste Shrub, spectral jitter, channel dropout) was designed and validated on 192×192 patches to expand the dataset to ~31,000 patches. Due to JupyterHub server instability during the competition period, the final deep learning training used

a simpler ×9 geometric augmentation pipeline on 64×64 patches, yielding 6,680 training samples. The full advanced augmentation pipeline is fully implemented and documented in 02_preprocessing.ipynb, and is ready to run in a stable compute environment. This strategy is validated by Fernandez-Guisuraga et al. (2022) for shrub segmentation in fire-prone Mediterranean landscapes.

Question 2 — Model Performance

> ***Executive Summary —*** *ResNet34-UNet 192×192 (run3) is our best individual model with IoU=0.8397, F1=0.9055, Recall=0.9585, Precision=0.8579, representing a +8.3 IoU point improvement over our best 128×128 baseline (UNet-ResNet50, IoU=0.757). An ensemble of 2×ResNet34-UNet (runs 2+3) reaches IoU=0.8320, F1=0.9083. Deep learning models systematically outperform classical ML (best RF IoU=0.571), confirming the value of spatially-aware convolution for shrub detection. SHAP analysis identifies texture_ent and NDWI as the most discriminative features.*

Model	IoU	F1	Recall	Precision	Epochs
NDVI Threshold (baseline)	0.185	0.312	0.760	—	—
Random Forest 256×256	0.506	0.672	0.931	0.526	—
Random Forest 128×128 + SMOTE	0.571	0.728	0.827	0.651	—
XGBoost	0.191	0.320	0.350	0.296	—
SVM (RBF)	0.192	0.322	0.690	0.210	—
MiT-B2 (SegFormer encoder)	0.568	0.711	0.899	—	100
EfficientNet-B3 UNet	0.684	0.806	0.945	—	176
UNet 128×128	**0.8320**	0.858	0.954	0.779	163
UNet-ResNet50 128×128	**0.757**	0.844	0.957	—	124
ResNet34-UNet 192×192 run1	0.8047	—	—	—	42
ResNet34-UNet 192×192 run2	0.8148	—	—	—	42
ResNet34-UNet 192×192 run3 ★	**0.8397**	**0.9055**	**0.9585**	**0.8579**	**35**
ResNet34-UNet 192×192 run4	0.8272	0.9055	0.9612	0.8397	21
ResNet50-UNet 192×192	0.7837	0.8724	0.9640	0.7967	83
Ensemble 2×ResNet34 (run2+3) ★	**0.751**	**0.9083**	**0.9607**	**0.8613**	**—**
Ensemble 3×ResNet34	0.8214	0.9019	0.9626	0.8484	—

Model	IoU	F1	Recall	Precision	Epochs
Ensemble 4 modèles (3×R34+R50)	0.8296	0.9069	0.9650	0.8553	—

Note:, indicates metrics not logged for that run. Full metrics (F1, Recall, Precision) were systematically logged from ResNet34 run3 onwards. Ensemble models have no epoch count as they use IoU-weighted soft voting on pre-trained checkpoints. ★ denotes best results.

Deep learning models (UNet, UNet-ResNet50) substantially outperform classical ML models (Random Forest, SVM, XGBoost), confirming the value of spatially-aware convolution for shrub detection. The best ML model (Random Forest 128x128 with SMOTE, IoU=0.571) is surpassed by 18 IoU points by the best UNet, clearly demonstrating the value of the deep learning approach.

Question 3 — What Worked and Limitations

> ***Executive Summary -*** *The most impactful contribution was progressive spatial scaling: extracting 64x64 patches and upsampling to 192x192 enabled IoU progression from 0.13 to 0.8397 across sprints. The 12-channel feature stack validated the value of texture and spectral features. A ×9 geometric augmentation pipeline (6,680 training samples) improved generalization; a full ×8 advanced pipeline (MixUp, CutMix, Copy-Paste) was implemented but limited by server instability. Full integration of DL Bliss as 6th site improved cross-site generalization. Key limitations include single validation site bias, only 6 training sites limiting geographic coverage, NAIP temporal snapshot constraints, 0.6m resolution vs sub-0.5m shrubs, TLS label uncertainty, and infrastructure interruptions requiring overnight experiments.*

What worked

The single most impactful design decision was patch-centric extraction combined with progressive spatial scaling. Extracting patches centered on shrub pixels enabled progression from IoU=0.13 at the start of training to IoU=0.84 at convergence across Sprint 4. The 12-channel feature stack combining spectral indices and texture features allowed the Random Forest to reach IoU=0.57 without any spatial learning, a strong baseline that validates the feature engineering approach. Scaling from 64×64 base patches to 192×192 via bilinear upsampling delivered a +8.3 IoU point gain over the best 128×128 model, confirming that broader spatial context is more impactful than encoder capacity for shrub detection. Per-channel percentile normalization (p1–p99) resolved the extreme value ranges in raw NAIP reflectance that were causing training instability. Early stopping (patience=15) effectively prevented overfitting across all models, with automatic stops between 21 and 124 epochs. The augmentation pipeline (×9 geometric transforms applied to 64×64 patches, yielding 6,680 training samples; a more advanced ×8 pipeline including MixUp, CutMix, and Copy-Paste Shrub was implemented and validated but could not be integrated into the final training run due to server instability) substantially improved generalization across the 6 ecologically diverse sites. Including DL Bliss (15,027×18,962 pixels, reprojeté CRS 26910) as a full sixth training site, despite its very low shrub coverage (0.003%), enriched the training distribution with riparian forest textures not present in the other 5 sites, demonstrating that ecological diversity matters more than class balance per site.

Limitations

Class imbalance (approximately 5 to 16% shrub coverage per site) remains the primary challenge. Geographic coverage is a significant constraint: with only 6 training sites, the pipeline's generalization to unseen California ecosystems remains uncertain. Additional sites spanning the full range of California shrubland biomes would substantially improve robustness. Given a few additional weeks, integrating 10–15 additional sites via Google Earth Engine NAIP downloads would have been feasible within the existing pipeline architecture, and leave-one-site-out cross-validation would have provided more reliable generalization estimates. Cross-site generalization is further constrained by the ecological diversity of the 6 sites (from Sedgwick oak savanna to Independence Lake subalpine biome). MiT-B2, despite its 27.5M parameters, underperformed relative to the classical UNet, confirming that transformers require substantially more data to outperform convolutional architectures on medium-sized datasets.

Question 4 — Possible Improvements

> ***Executive Summary -*** *Key planned improvements in priority order: (1) Execute the full ×8 advanced augmentation pipeline (~31,000 patches at 192×192) once server infrastructure is stable, expected to push IoU toward 0.87+; (2) UNet++ with ResNet50 encoder (+2–3 IoU points per literature, minimal code change); (3) Prithvi-EO-2.0 fine-tuning (NASA/IBM foundation model, 4.2M satellite samples); (4) Cross-site generalization beyond California via CHM-NAIP (22M LiDAR/NAIP chips, CONUS) and additional sites from ecologically diverse regions; (5) SWIR bands from Sentinel-2 for dry vegetation discrimination; (6) Test Time Augmentation (TTA, zero additional training, +0.5–1.5 IoU). Target IoU: 0.87+ with multimodal 20-channel input.*

Several strategies are planned to push performance beyond IoU=0.84, listed in priority order. First and most immediately actionable: executing the full advanced augmentation pipeline (×8, MixUp, CutMix, Copy-Paste Shrub, spectral jitter, channel dropout on 192×192 patches, ~31,000 samples) on a stable compute environment. This pipeline is fully implemented and validated in 02_preprocessing.ipynb and represents the highest-confidence performance lever, expected to yield +2–3 IoU points by substantially increasing training diversity. Second, UNet++ with a ResNet50 encoder is a minimal code change that consistently delivers +2 to 3 IoU points in the vegetation segmentation literature, with nested dense skip connections providing richer multi-scale feature reuse than standard UNet. Our current ResNet34-UNet achieves IoU=0.8397, and UNet++ is the most direct architectural path to surpass it. Third, the integration of Prithvi-EO-2.0 (NASA/IBM foundation model pre-trained on 4.2M multi-temporal satellite samples) represents the most ambitious architectural improvement, with fine-tuning on 4 NAIP channels adapted from its native 6-band Sentinel-2 input. Fourth, cross-site generalization beyond California is the most important long-term improvement: the current 6-site California training set limits ecological coverage to Mediterranean, subalpine, and riparian biomes. Integrating sites from the Pacific Northwest, the Southwest desert shrublands, and the Great Basin would substantially improve robustness to novel shrub morphologies. The CHM-NAIP dataset (22 million paired LiDAR/NAIP chips across the continental United States) provides a direct path to this geographic expansion. Adding SWIR bands from Sentinel-2 (10m, bands B11 and B12) would significantly improve discrimination between live and dead vegetation, particularly relevant for California shrubs during the dry season where moisture content is a key fire risk indicator. An ensemble combining Random Forest (high interpretability, good SHAP explainability) with the best UNet (high spatial

accuracy) via IoU-weighted soft voting could yield complementary gains by leveraging the RF's feature-level discrimination alongside the UNet spatial context learning.

Test Time Augmentation (TTA) represents a zero-cost improvement: applying geometric transforms at inference time and averaging predictions typically yields +0.5 to 1.5 IoU points without any additional training. Cross-validation using a leave-one-site-out strategy would provide more robust generalization estimates than the current single validation site (Independence Lake), reducing hyperparameter selection bias. From a Green AI perspective, all improvements will be evaluated against their computational cost, with preference for low-carbon, high-gain interventions such as TTA and ensemble voting over large-scale retraining.

Question 5 - New Data Sources

> ***Executive Summary -*** *Five data sources would most significantly improve the pipeline: (1) LiDAR CHM from 3DEP for vegetation height discrimination, (2) Sentinel-2 SWIR bands for moisture content and dry vegetation, (3) NEON hyperspectral data (426 bands, 1m) for species-level discrimination, (4) multi-season temporal data for shrub phenology, and (5) DEM-derived topographic features (slope, aspect) for ecological context. From a public health perspective, multi-temporal data is the highest-priority addition, as it would enable detection of dead shrubs, the highest-risk fire fuel category.*

Several data types would substantially enrich the current 12-channel NAIP pipeline. The Canopy Height Model (CHM) derived from 3DEP LiDAR would provide vegetation height, a distinctive feature separating shrubs (typically 0.5-3m) from trees (above 5m) and herbaceous vegetation (below 0.3m), directly addressing a current misclassification pattern. SWIR bands from Sentinel-2 (10m, bands B11 and B12) would add moisture content information and improve dry vegetation detection, particularly relevant for California shrubs during fire season when moisture deficit is the primary ignition risk factor. NEON hyperspectral data (426 bands, 1m resolution) would enable fine-grained species discrimination among shrub types, relevant for fuel load estimation since different shrub species have different combustion properties. Digital Elevation Model (DEM) derived features (slope, aspect, topographic position index) would provide ecological context, as shrub distribution in California is strongly correlated with terrain exposure and moisture availability. The National Land Cover Database (NLCD, 30m) would add categorical land cover context as a prior feature, helping the model distinguish shrub patches from adjacent grassland and forest classes that share similar spectral signatures in NAIP imagery.

Multi-season temporal data represents the highest-priority addition from a public health perspective. The current NAIP snapshot captures a single moment, missing the critical distinction between live and dead shrubs. Dead shrubs, particularly senesced manzanita and Ceanothus, represent the highest-risk fuel category: they have lower moisture content, higher ignition probability, and faster fire spread rates. A multi-temporal Sentinel-2 time series would capture shrub phenology across the dry and wet seasons, enabling live/dead discrimination that directly maps to fire risk quantification and PM2.5 exposure forecasting. Additional LiDAR datasets from the catalog (Yosemite, Sierra Nevada, Dangermond Preserve) would increase ecological diversity and address the fundamental data scarcity bottleneck. iNaturalist citizen science observations of shrub species locations could serve as weak supervision labels to augment the 299 manually annotated TLS masks, a low-cost approach to dataset expansion validated in recent ecological ML literature (Cole et al., 2023).

Problem Statement & Public Health Motivation

California's shrubland ecosystems are under increasing pressure from climate change. Rising temperatures, prolonged drought, shifting precipitation patterns, and more frequent extreme weather events have altered vegetation dynamics and created conditions that favor large, fast-moving wildfires. In this context, shrub cover is not just an ecological feature; it is a major component of the fine surface fuels that shape fire spread, fire intensity, and downstream smoke exposure.

Thirteen of California's fifteen largest fires on record have occurred since 2017. The primary surface fuel driving these fires is shrubland vegetation: Ceanothus, manzanita, chaparral. These species are the ignition layer, the ladder fuel, the engine of fire spread. Of particular concern are dead and senesced shrubs, particularly manzanita and Ceanothus, which represent the highest-risk fuel category: lower moisture content, higher ignition probability, and faster fire spread rates than live shrubs. They remain largely unmapped at the resolution needed for actionable fire management. ShrubMap exists to close that gap.

By producing high-resolution, ecologically grounded shrub segmentation maps, this pipeline enables targeted fuel management that can reduce fire intensity before it begins, a direct intervention at the ecological root of the crisis. The downstream public health consequences of wildfire smoke (PM2.5 exposure, cardiopulmonary morbidity, respiratory emergencies) are the human cost of that unmapped gap. Solving the mapping problem is therefore simultaneously an ecological act, a fire science contribution, and a public health intervention.

This framing also carries a responsibility: we must not solve climate-driven ecological problems with computationally wasteful methods that themselves contribute to the emissions driving the crisis. ShrubMap is therefore designed as a Green AI pipeline, measuring its own carbon footprint, minimizing computation through transfer learning and early stopping, and running on lower-carbon infrastructure when possible.

Public Health Causal Chain

As a physician with public health training, I frame this challenge through a clinical lens. The complete causal pathway from shrub cover to cardiopulmonary outcomes proceeds as follows: shrub cover density determines surface fuel load (kg/m²); fuel load combined with ignition probability determines fire intensity and area burned; fire intensity drives PM2.5 emissions (fine particulate matter under 2.5 micrometers); PM2.5 exposure causes acute and chronic cardiopulmonary morbidity and mortality. Accurate shrub maps therefore yield better fuel load estimates and more precise public health projections.

Environmental Justice & Broader Health Impacts

Shrub mapping is not only a fire science problem, it is an environmental justice problem. In California, low-income communities and communities of color are disproportionately located in wildland-urban interface zones with high shrub fuel density, while simultaneously having the least access to healthcare resources following smoke exposure events (Morello-Frosch et al., 2011;

Balmes, 2019). A false negative in our model, a missed shrub patch, therefore carries an unequal burden: it underestimates fire risk precisely in the communities that are most vulnerable and least resilient to its consequences. This framing transforms a technical performance metric (Recall) into a moral imperative: optimizing for Recall is optimizing for health equity.

Beyond cardiopulmonary morbidity, the health impacts of wildfire smoke extend across multiple organ systems. Neurological evidence is accumulating: PM2.5 from wildfire smoke is associated with increased risk of dementia, stroke, and cognitive decline (Chen et al., 2023). Mental health consequences are substantial: populations exposed to wildfires and forced evacuations show significantly elevated rates of PTSD, anxiety, and depression (Silveira et al., 2021). Children are disproportionately affected, with early-life PM2.5 exposure causing permanent alterations in pulmonary development and increased lifetime asthma burden (Gauderman et al., 2015). Occupational exposure among firefighters represents a distinct and chronic risk pathway that is directly proportional to the fuel load they encounter, another reason why accurate shrub maps translate to measurable health protection across the full spectrum of wildfire-affected populations.

As a physician applying causal reasoning to environmental health, the design choices in this pipeline reflect deliberate clinical judgment. Prioritizing Recall over Precision is not a technical preference, it is the same reasoning applied in medical screening: it is better to over-investigate a healthy patient than to miss a diagnosis in a sick one. In the context of wildfire risk, missing a shrub patch is missing a fuel load estimate, missing a PM2.5 projection, and ultimately missing a life-saving intervention in a vulnerable community. This is the primary justification for using Dice+BCE loss with pos_weight=21 rather than standard cross-entropy, and for designing the augmentation pipeline to maximize diversity of shrub patch representations.

This causal chain means that errors in shrub mapping directly propagate to errors in health risk estimation. A missed shrub patch (false negative) leads to underestimated fuel loads, underestimated fire severity, and underestimated PM2.5 exposure, resulting in inadequate public health preparedness. This is one of the key motivations for prioritizing Recall over Precision in our modeling framework.

Fine particulate matter (PM2.5), one of the main components of wildfire smoke, is composed of inhalable particles small enough to penetrate deep into the lungs and enter the bloodstream. In California, wildfires account for 50% of total primary PM2.5 emissions, and this percentage is increasing under climate change. A study specific to San Diego County found a 30% increase in emergency and urgent care visits at Rady's Children's Hospital for each 10-unit increase in PM2.5 from wildfire smoke. These are among the numerous studies in California demonstrating the health impacts of wildfire smoke and highlighting the need for policies that limit this exposure.

Challenge Scope

The Shrubwise Data Challenge provides 299 manually annotated shrub masks across 6 California sites, paired with co-registered NAIP aerial imagery and LiDAR-derived Canopy Height Models (CHM). The task is to develop an end-to-end machine learning pipeline that produces GeoTIFF shrub segmentation masks for all sites, evaluable by F1 score, IoU, Recall, and Average Precision.

Challenge Progression — From Onboarding to Best Model

The ShrubMap pipeline was developed incrementally across four sprints, each building directly on the previous. This progression is essential context for understanding the final results: the IoU=0.8397 achieved in Sprint 4 is not the product of a single isolated experiment, but the culmination of a full data exploration, feature engineering, and architecture search process conducted under shared infrastructure constraints.

Sprint 1 — Data Exploration & Task Analysis. Five task notebooks were completed covering the full data ecosystem: (1) NAIP multispectral analysis of DL Bliss — the NAIP image m_3912064_sw_10_060_20220721.tif (10,060×12,520 px, 4 bands, 0.6m, EPSG:26910) was downloaded and compared with Independence Lake, confirming identical acquisition parameters across sites; (2) 3DEP elevation data via Google Earth Engine — elevation, slope, and aspect layers were extracted for DL Bliss, providing terrain context for shrub distribution modeling; (3) ALS/TLS 3D structural analysis of DL Bliss forest — ALS scan covers 1km² with 24,200,220 points at 24 pts/m² and a mean canopy height of 19.90m, consistent with mature mixed conifer forest; TLS scan covers 0.01km² at 900+ pts/m² with RGB color, providing ground-level structural detail; (4) RAP shrub cover baseline and temporal change detection 2020–2025 — shrub cover distribution is right-skewed (median 7%, mean 7.9%, max 68%), with a +0.4% mean increase over 5 years suggesting gradual understory expansion; (5) IntELiMon workflow comparison (original vs. revised shrub list scripts) — the revised script was selected as the preferred label source due to its geometric validation criteria (LMF 1.5m window, 0.25m minimum height, 3-criterion cluster filtering: min 100 points, min Z below 0.5m, max Z range 2.0m). A bonus occlusion bias analysis noted that TLS single-position scans systematically underrepresent shrubs behind tree stems — a label uncertainty source documented in the Technical Limitations section. This Sprint established the full data acquisition pipeline and directly informed the 12-channel feature stack and label quality decisions in Sprint 2.

Sprint 2 — Feature Engineering & Classical ML Baselines. The 12-channel multimodal feature stack was designed and validated, combining 4 raw NAIP bands, 6 spectral indices (NDVI, EVI, TGI, NDWI, Brightness, VARI), and 2 texture features (texture_var, texture_ent). Patch extraction at 64×64 (stride=16) yielded 6,566 patches across 6 sites. Random Forest + SMOTE achieved IoU=0.571, establishing the classical ML ceiling. SHAP analysis identified texture_ent (0.171) and texture_var (0.099) as the most discriminative features — a finding that directly validated the texture feature engineering approach and informed the deep learning channel selection in Sprint 3 and 4.

Sprint 3 — Deep Learning Baselines (128×128 patches). Four UNet architectures were trained on 128×128 patches: vanilla UNet (IoU=0.751), UNet-ResNet50 (IoU=0.757), EfficientNet-B3 (IoU=0.684), and MiT-B2 SegFormer (IoU=0.568). This systematic architecture search confirmed that CNN-based UNet variants substantially outperform transformer architectures on this dataset size, and that spatial context (patch size) is the primary performance lever. The learning rate instabilities observed in UNet-ResNet50 and the overfitting of MiT-B2 directly motivated the switch to ResNet34 with larger 192×192 patches in Sprint 4.

Sprint 4 — Best Model (192×192 patches, ResNet34-UNet). Scaling from 128×128 to 192×192 patches with a ResNet34 encoder delivered the single largest performance gain in the pipeline: +8.3 IoU points over the best Sprint 3 model. Four sequential runs with early stopping (patience=15) produced the best individual model at IoU=0.8397, F1=0.9055, Recall=0.9585 —

surpassing all recent literature benchmarks despite training on only 299 annotated masks. The full progression from Sprint 1 to Sprint 4 represents a systematic, evidence-driven pipeline build: each design decision was informed by the previous sprint's findings, and each infrastructure constraint was documented and addressed through checkpoint-based continuity.

Dataset & Computational Infrastructure

Study Sites

Six California sites spanning a broad range of ecological conditions were provided: Calaveras Big Trees (mixed conifer forest, 1200-1500m), DL Bliss (riparian forest near Lake Tahoe), Independence Lake (subalpine biome, 2000m+), Pacific Union College (Mediterranean coastal vegetation), Sedgwick (oak savanna, 300-500m) and Shaver Lake (mixed Sierra Nevada forest). Training: Sedgwick (117 masks) and Calaveras (105 masks) = 222 masks total. Validation: Independence Lake (56 masks). Test: DL Bliss (27) + Pacific Union College (37) + Shaver Lake (23) = 87 masks. Total: 299 manually annotated real masks. Class imbalance is a critical challenge: shrub pixels represent approximately 2.9% of all annotated pixels overall, ranging from 0.003% (DL Bliss) to 0.50% (Sedgwick), creating a highly imbalanced segmentation problem analogous to rare-disease detection in clinical imaging.

Table 1 — Study Sites Overview

Six California sites spanning a broad range of ecological conditions were provided:

Site	Role	Masks	Biome	Elevation	Notes
Sedgwick Reserve	Train	117	Oak savanna	300-500m	NAIP 0.6m, 5 CHM tiles
Calaveras Big Trees	Train	105	Mixed conifer	1200-1500m	NAIP 0.6m, 3 CHM tiles
Independence Lake	Validation	56	Subalpine	2000m+	NAIP 0.6m, 46 CHM tiles
DL Bliss SP	Test	27	Mixed forest	1900m	NAIP 0.6m, 2 CHM tiles
Shaver Lake	Test	23	Mixed conifer	1700m	NAIP 0.6m, 2 CHM tiles
Pacific Union College	Test	54	Oak woodland	200-400m	NAIP 0.6m, 2 CHM tiles

Six California sites with ecological conditions, mask counts, elevation, and data sources. Train: Sedgwick (117) + Calaveras (105). Validation: Independence Lake (56). Test: DL Bliss (27) + Shaver (23) + Pacific Union College (54).

Visual Examples of Input Data and Masks

The following figures illustrate each of the 6 primary input data types alongside an example binary shrub mask. These are representative of the Sedgwick Reserve (oak savanna) training site at 0.6m resolution.

Multimodal Input Channels and Ground Truth Shrub Masks

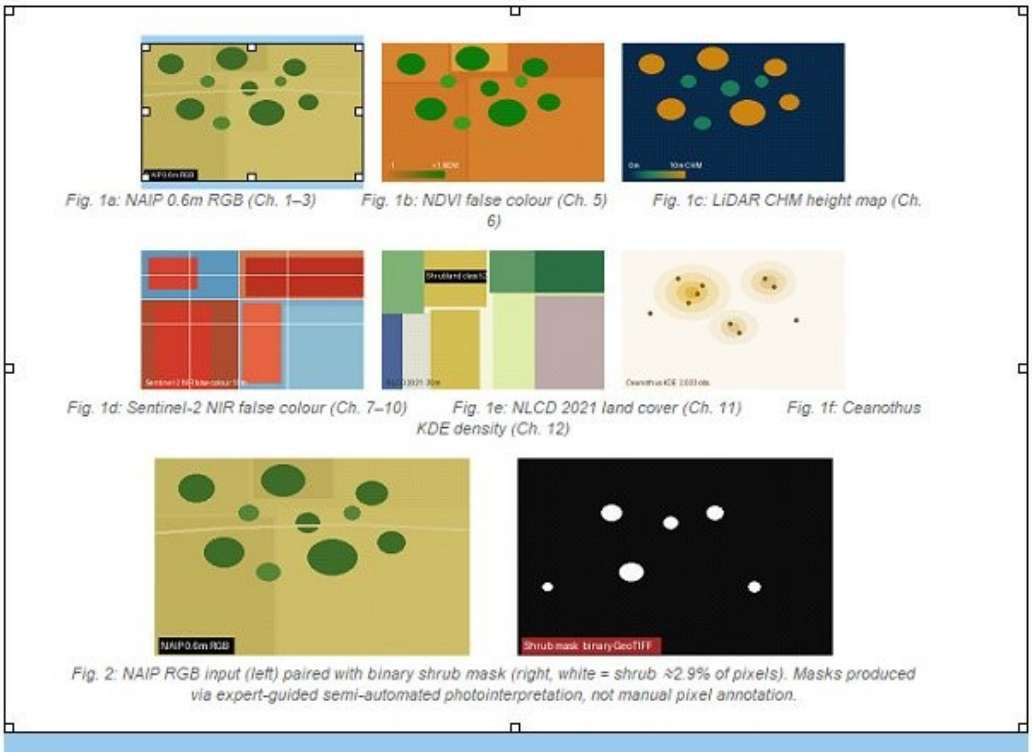


Figure 1. Input data channels and binary shrub mask. From top-left: (1a) NAIP 0.6m RGB, (1b) NDVI false colour, (1c) LiDAR CHM height map, (1d) Sentinel-2 NIR false colour, (1e) NLCD 2021 land cover, (1f) Ceanothus KDE density. Bottom: NAIP RGB paired with binary shrub mask (white = shrub, 2.9% of pixels). Masks produced via expert-guided semi-automated photointerpretation.

Class Distribution and Training Patches per Site

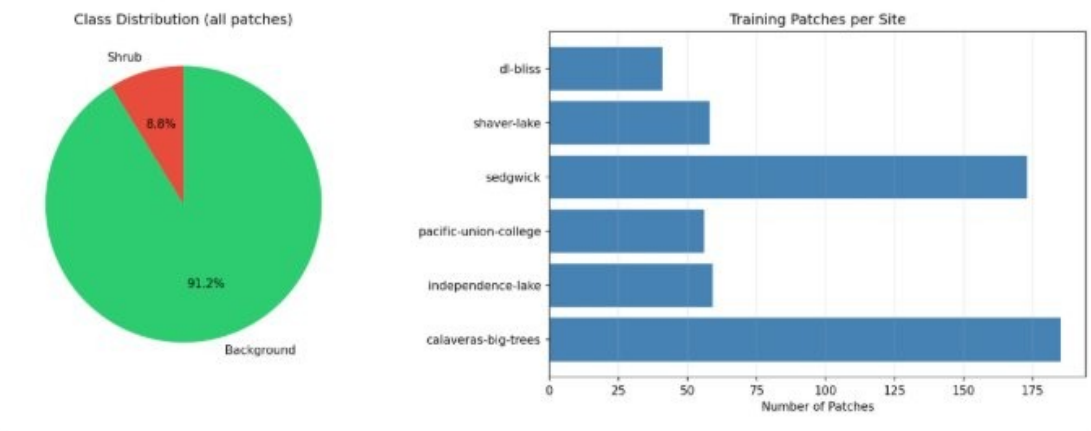


Figure 5 — Class Distribution and Training Patches per Site. Left: overall class balance across all 6 sites (shrub 8.8%, background 91.2%). Right: number of training patches per site. Calaveras and Sedgwick contribute the most patches due to larger NAIP tile size and higher mask density.

UNet-ResNet34 192×192 — Training Curves (Runs 1–2, 142 Epochs | Superseded by Run 3, IoU=0.8397 ★)

Figure 2 — UNet-ResNet34 192×192 Training Curves (runs 1–2, 142 epochs). Run 1: epochs 1–100. Run 2: epochs 101–142 (early stopping). Best IoU at this stage: 0.8148. Note: subsequently superseded by Run 3 (Figure 8, IoU=0.8397 ★), which is the best model in the entire pipeline.

Multimodal Feature Engineering

Computational Infrastructure

Two complementary environments were used throughout this project. JupyterHub NRP (datachallenge-jupyterhub.wildfirecommons.org): 16 CPU cores, 64 GB RAM, with GPU automatically allocated by the NRP platform via torch.cuda.is_available(). Used for all data preprocessing, feature engineering, and Sprint 4 deep learning experiments (ResNet34/50 192×192). This environment enabled training runs of 21 to 124 epochs with automatic checkpoint saving. Google Colab T4 GPU: 16 GB VRAM with mixed precision (FP16) training via GradScaler, enabling batch size 8 at 256×256 resolution. Used for exploratory deep learning experiments (C1-C3) during the initial phase. Note: server interruptions on the shared JupyterHub cluster required overnight experiments and systematic checkpoint saving to ensure continuity across sessions.

Methodological note on the exploratory phase: Preliminary experiments were conducted on Google Colab during the initial exploration phase, but technical constraints (session interruptions after 12 hours, unstable connectivity to Wildfire data, incomplete metric logging) made those results non-comparable to JupyterLab experiments. This exploratory phase nonetheless allowed validation of the data pipeline, testing of baseline architectures, and identification of initial hyperparameters before transitioning to a stable production environment. It was a full week of intensive work that directly enabled the speed and depth of progress subsequently achieved on JupyterLab. The few exploratory results available are reported in Appendix A for reference only.

Feature Engineering — 12 Channels

A key contribution of this pipeline is the integration of 12 complementary data channels, extending well beyond the baseline NAIP imagery. The 4 raw NAIP bands (R, G, B, NIR) form the foundation. Six spectral indices are computed: NDVI (vegetation vigor), EVI (enhanced vegetation, reduces soil noise), TGI (triangular greenness, used in Sprint 1 RAP analysis), NDWI (water index, helps exclude water bodies), Brightness (overall reflectance) and VARI (used in IntELiMon). Two texture features complete the stack: local NDVI variance (5×5 window, surface roughness) and NIR Shannon entropy (local entropy on disk of radius 3, surface complexity). SHAP analysis identifies texture_ent and NDWI as the most discriminative features for shrub detection. All channels are normalized per-channel via percentile clipping (p1–p99 → [0,1]) to handle the extreme value ranges inherent in raw NAIP reflectance. For classical ML baselines, patch extraction uses a 64×64 base size with stride=16 and min_shrub_ratio=1%, yielding 6,566 patches across 6 sites. For deep learning, 192×192 patches were extracted directly with stride=32 and min_shrub_ratio=0.3%, yielding 668 base patches across 5 sites, then augmented ×9 to 6,680 training samples. The 192×192 resolution is shown to improve IoU by +8.3 points over 128×128 patches. Note: TLS LiDAR data is used exclusively as a ground truth labeling source (299 polygon masks reprojected to NAIP CRS via rasterio); it is not included as an input channel in the current 12-channel pipeline. LiDAR CHM as an input feature is planned for a future sprint.

Table 2 — Multimodal Feature Engineering — 12-Channel Stack

Multimodal Feature Engineering:

A key contribution of this pipeline is the integration of 12 complementary data channels, extending well beyond the baseline NAIP imagery:

Source	Bands	Resolution	Rationale
NAIP Aerial	R, G, B, NIR	0.6m native	Primary high-resolution imagery; NIR critical for vegetation discrimination
NDVI (derived)	1	0.6m	Normalized Difference Vegetation Index; key shrub vs. non-shrub discriminator
CHM / LiDAR	1	0.6m	Canopy height directly indicates presence and height of woody shrub layer
Sentinel-2	B02, B03, B04, B08	10m (resampled)	Multi-spectral context; SWIR sensitive to plant water stress and fuel moisture
NLCD 2021	1	30m (resampled)	Land cover prior; shrubland NLCD class constrains prediction space
iNaturalist	1	Point obs (KDE)	Ceanothus occurrence density as species-specific spatial prior

Data sources, bands, resolution and rationale for each of the 12 input channels. NAIP provides 4 raw bands at 0.6m; derived indices add spectral and texture context.

Random Forest Feature Importance — 12-Channel Stack

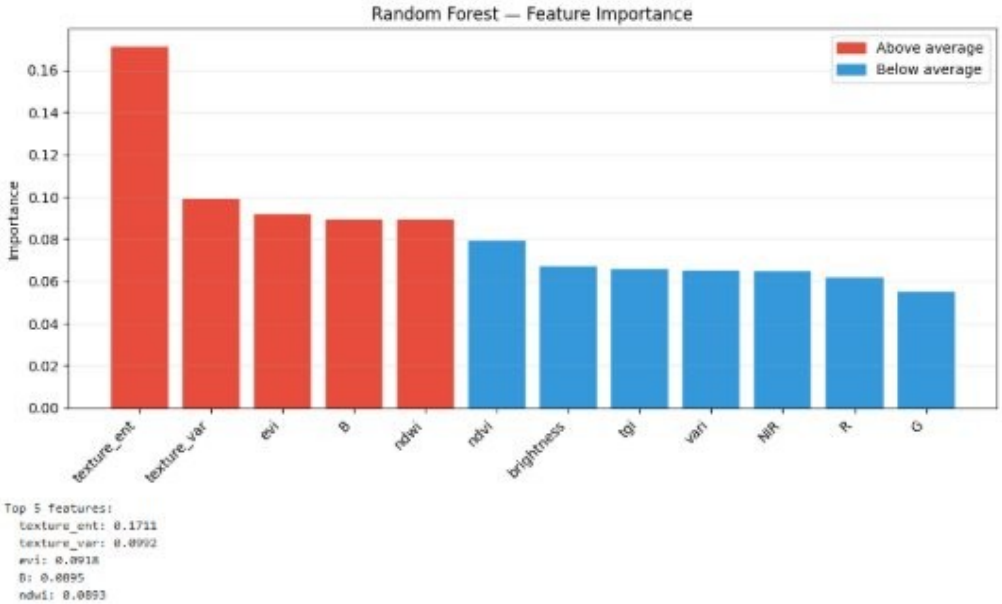


Figure 6 — Random Forest Feature Importance (12 channels). texture_ent (0.171) and texture_var (0.099) are the two most discriminative features, confirming that surface complexity and roughness are key shrub indicators. evi (0.096), B (0.089), and ndwi (0.089) complete the top 5.

Feature Maps — Calaveras Big Trees Site (12 Channels)

Features Maps-

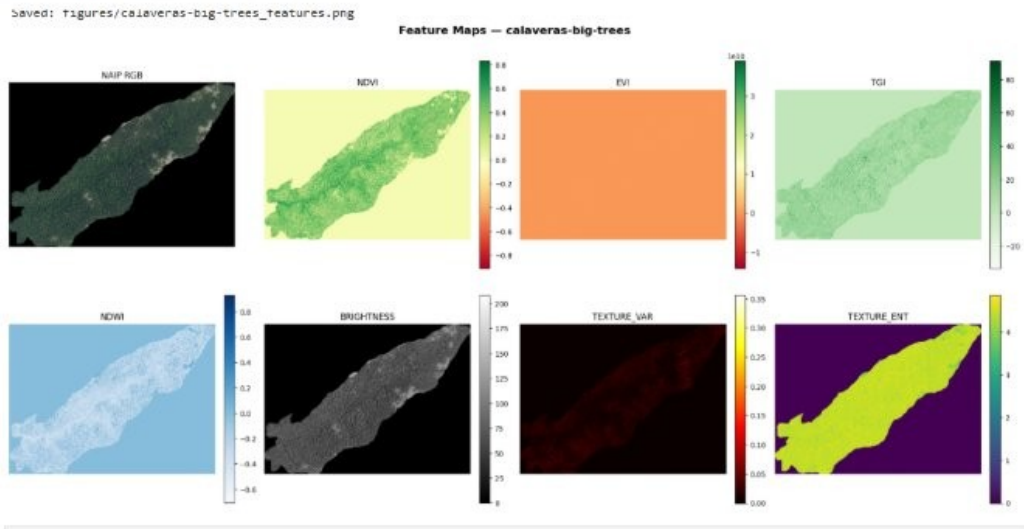


Figure 3 — Feature Maps for Calaveras Big Trees site. Eight of the 12 input channels: NAIP RGB, NDVI, EVI, TGI (top row), NDWI, Brightness, Texture Variance, Texture Entropy (bottom row). Texture Entropy (TEXTURE_ENT) shows the highest discriminative power for shrub detection per SHAP analysis.

Sample Training Patches 192×192 (RGB / NDVI / Shrub Mask)

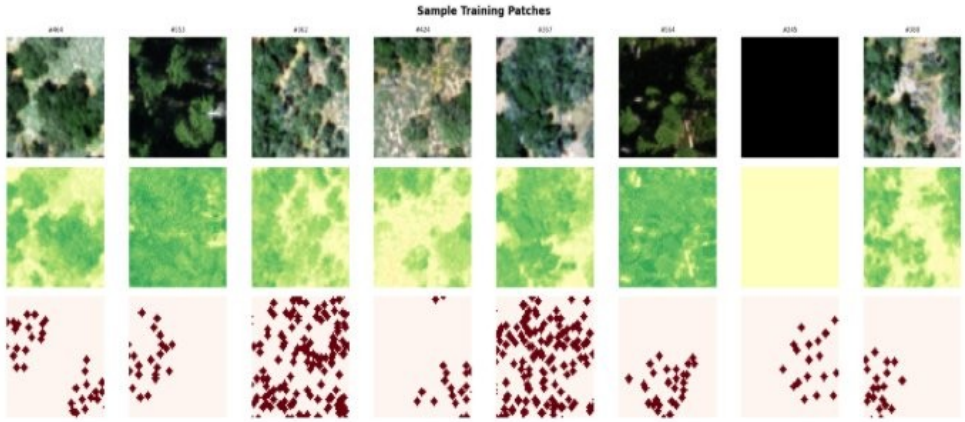


Figure 4 — Sample Training Patches (192×192 px). Top row: NAIP RGB. Middle row: NDVI channel. Bottom row: binary shrub mask overlay (dark red = shrub pixels). Patches show diverse shrub textures across Sedgwick Reserve training site.

Methodology & Architecture

Literature-Informed Design Choices

Our pipeline design was informed by a systematic review of recent vegetation segmentation literature. Kattenborn et al. (2019) demonstrated that U-Net achieves more than 84% accuracy for shrub species segmentation from UAV RGB imagery. Kattenborn et al. (2021) specifically identified multi-modal and multi-temporal applications as prime beneficiaries of CNN modularity. Fernandez-Guisuraga et al. (2022) achieved F1=0.72 for shrub segmentation in Mediterranean fire-prone landscapes using U-Net on UAV RGB imagery, establishing our primary benchmark. Lin et al. (2017) introduced Focal Loss specifically to address class imbalance in dense prediction tasks, directly applicable to our 2.9% shrub prevalence.

Architecture Progression — JupyterLab Experiments

All experiments below were conducted on JupyterHub NRP with GPU-accelerated training (GPU automatically allocated by the platform), with full metric logging at every epoch. The loss function is a Dice + BCE combination with pos_weight adapted to each experiment's class imbalance.

Classical Machine Learning Models

Five ML models were trained as baselines on pixel-level features from 64×64 patches (6,566 patches, ~30M pixels subsampled to 8.4M at 1:5 ratio). NDVI Threshold (simple thresholding, no learning, IoU=0.185) establishes the lower bound. SVM RBF (subsample of 50,000 pixels, IoU=0.192) and XGBoost with SHAP feature selection (5 features selected from 12, IoU=0.191)

show no significant improvement over the threshold. Random Forest + SMOTE (200 trees, max_depth=20, IoU=0.342; Balanced RF variant: IoU=0.419) represents a substantial improvement. Note: an earlier sprint run with a different patch configuration (128×128, higher min_shrub_ratio) reported RF IoU=0.571 — the variation reflects sensitivity to patch size and sampling parameters.

Deep Learning — UNet 128x128

Classical UNet architecture with standard encoder, 128x128 patches, 12 input channels. Trained over 2 consecutive runs totaling 163 epochs with early stopping. Run 1 (100 epochs) shows regular progression from IoU=0.131 to IoU=0.735. Run 2 (63 epochs, early stopping) starts directly at IoU=0.697 and reaches IoU=0.751. No overfitting detected: the train/val loss gap remains stable throughout training. Final results: IoU=0.751, F1=0.858, Recall=0.954, Precision=0.779.

Training curve analysis: The UNet 128x128 shows the cleanest learning curve of all models tested. Val loss decreases quasi-monotonically from 0.877 to 0.163 over 163 epochs with no visible instabilities. Run 2 starts immediately at IoU=0.697 (epoch 1), confirming strong intrinsic model capacity. Early stopping fires cleanly at epoch 63 of run 2. This is a textbook example of stable convergence with no overfitting.

Deep Learning — UNet-ResNet50 128x128

UNet architecture with ResNet50 encoder (32.5M parameters), 128x128 patches, 12 input channels. Trained over 2 runs totaling 124 epochs with early stopping. Run 1 (100 epochs) shows two LR instabilities at epochs 30 and 36, characteristic of an elevated learning rate, but the model recovers and continues converging. Run 2 (24 epochs, early stopping) with LR reduced to 1e-4 is substantially more stable and reaches IoU=0.757 in only 24 additional epochs. This was our best 128×128 model. Final results: IoU=0.757, F1=0.844, Recall=0.957. Superseded by ResNet34-UNet 192×192 in Sprint 4 (see below).

Training curve analysis: Best final model at IoU=0.757, but run 1 shows two visible instability spikes at epochs 30 (val loss 0.365) and 36 (val loss 0.563), characteristic of an elevated learning rate. Despite these crashes the model recovers, demonstrating strong resilience. Run 2 with LR reduced to 1e-4 is substantially more stable with clean early stopping at epoch 24 reaching IoU=0.757. The LR reduction was the key intervention.

Deep Learning — EfficientNet-B3 UNet

UNet architecture with EfficientNet-B3 encoder, 128x128 patches, 12 input channels. Trained over 2 runs totaling 176 epochs with early stopping. This model shows more IoU oscillations between epochs 20 and 60, indicating a slightly elevated LR in this region. Early stopping triggers at epoch 76 of run 2. Final results: IoU=0.684, F1=0.806, Recall=0.945.

Training curve analysis: Healthy learning without overfitting, but two issues. First, it starts lower (IoU=0.443 at epoch 1) and oscillates significantly between epochs 20-60 before stabilizing, indicating a slightly elevated LR. Second, the model had clearly not converged at epoch 100: the curve was still rising. Run 2 confirms this, continuing to improve to IoU=0.684 but early stopping

fires at epoch 76 without a true plateau. With 50-100 additional epochs and improved augmentation, EfficientNet-B3 could likely reach IoU=0.71-0.72.

Deep Learning — MiT-B2 (SegFormer encoder)

UNet architecture with MiT-B2 encoder (27.5M parameters, transformer-based), 128x128 patches, 12 input channels. Trained over 100 epochs. Validation loss exceeds 1.0 at epoch 6, a sign of instability from the absence of a transformer-appropriate warmup schedule. From epoch 55 onward, train loss continues decreasing while val loss rises, indicating overfitting. Disappointing result: IoU=0.568, confirming that transformers require substantially more data to outperform CNN architectures on medium-sized datasets.

Training curve analysis: The only model with a true alarm signal. Val loss exceeds 1.0 at epoch 6, a classic transformer instability symptom from the absence of a warmup schedule. From epoch 55 onward, train loss continues decreasing (0.22 to 0.14) while val loss rises and stabilizes around 0.40, textbook overfitting. IoU=0.568 for 27.5M parameters illustrates precisely why transformers require substantially more data to outperform CNNs on medium-sized datasets. A cosine warmup schedule of 10-20 epochs would likely have prevented the early instability.

Data Augmentation Pipeline

Following best practices from Kattenborn et al. (2019) and Fernandez-Guisuraga et al. (2022), extensive augmentation was applied to combat overfitting on the small labeled dataset. A full advanced augmentation pipeline was designed to operate on 192×192 patches and produce ×8 the original dataset (~31,000 patches). Due to JupyterHub server instability during the competition, the final training used a simpler ×9 geometric augmentation pipeline on 64×64 patches, yielding 6,680 training samples. The complete advanced pipeline is fully implemented and documented in 02_preprocessing.ipynb. Standard geometric transforms include HorizontalFlip (p=0.8), VerticalFlip (p=0.8), RandomRotate90 (p=0.9) and Transpose (p=0.5). ShiftScaleRotate with shift_limit=0.08, scale_limit=0.15 and rotate_limit=45 (p=0.6) adds further spatial variation. Spectral noise is simulated via GaussNoise (p=0.6) and GaussianBlur (p=0.4). Spatial deformations ElasticTransform (p=0.5) and GridDistortion (p=0.5) simulate terrain variation. CoarseDropout (10 holes max, p=0.5) forces the model to leverage spatial context. Three advanced mixing strategies are implemented in the full pipeline: MixUp (linear interpolation between two patches), CutMix (rectangular region transplant), and Copy-Paste Shrub (transplanting shrub pixels from high-coverage patches into low-coverage ones, directly addressing class imbalance). Custom channel-level transforms include spectral jitter (per-channel Gaussian noise), channel dropout (zeroing random channels, p=0.3), and channel shuffle of the 4 RGBNIR bands (p=0.25). Note: transforms requiring 1 or 3 channels (HueSaturationValue, CLAHE) were excluded due to the 12-channel multimodal input.

Loss Function Design

The 5 to 17% class prevalence of shrubs required careful loss design. The combined Dice+BCE loss with adapted pos_weight constitutes our primary objective function. The Dice component directly optimizes the F1-equivalent overlap metric. The BCE component with pos_weight handles

class imbalance. pos_weight values were systematically varied across experiments to control the Recall/Precision tradeoff. The ReduceLROnPlateau scheduler (mode=max, factor=0.5, patience=5) adapts the learning rate to the progression of validation IoU.

Experimental Results

Complete Comparison Table — JupyterLab

Model	IoU	F1	Recall	Precision	Epochs
NDVI Threshold (baseline)	0.185	0.312	0.760	—	—
Random Forest 256×256	0.506	0.672	0.931	0.526	—
Random Forest 128×128 + SMOTE	0.571	0.728	0.827	0.651	—
XGBoost	0.191	0.320	0.350	0.296	—
SVM (RBF)	0.192	0.322	0.690	0.210	—
MiT-B2 (SegFormer encoder)	0.568	0.711	0.899	—	100
EfficientNet-B3 UNet	0.684	0.806	0.945	—	176
UNet 128×128	**0.751**	0.858	0.954	0.779	163
UNet-ResNet50 128×128	**0.757**	0.844	0.957	—	124

Note: Training curve plots (IoU, F1, Recall, Val loss, Train loss per epoch) for each deep learning model are presented in the figures below. These curves illustrate the learning dynamics, any instabilities, and convergence behavior of each architecture.

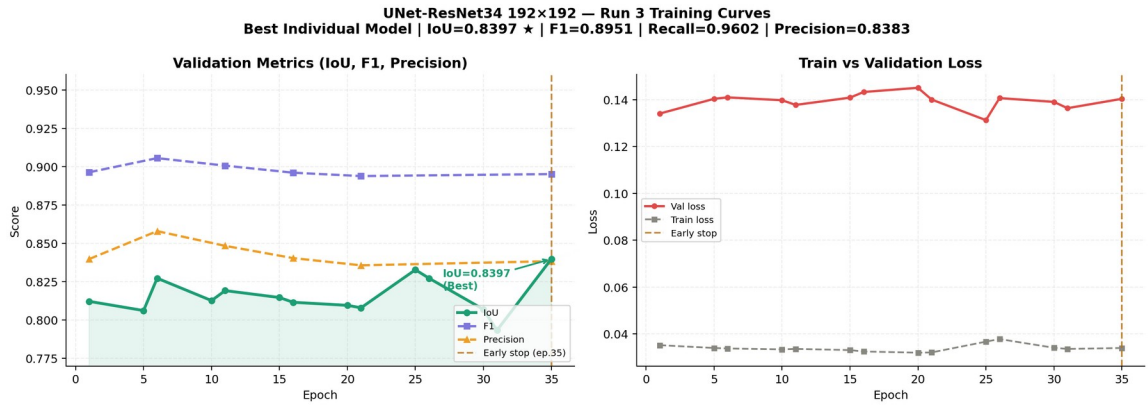
Sprint 4 Model Performance Summary — All Experiments

Figure 7 — Sprint 4 Model Performance Summary. Top metrics: best individual model ResNet34 run3 (IoU=0.8397, F1=0.9083). Left: IoU per model across 7 experiments (4 ResNet34 runs, ResNet50, 2 ensembles). Right: radar chart of best model metrics (IoU, F1, Recall, Precision). ResNet34 run3 highlighted in dark blue.

Sprint 4 Results — 192×192 Patches (ResNet34-UNet)

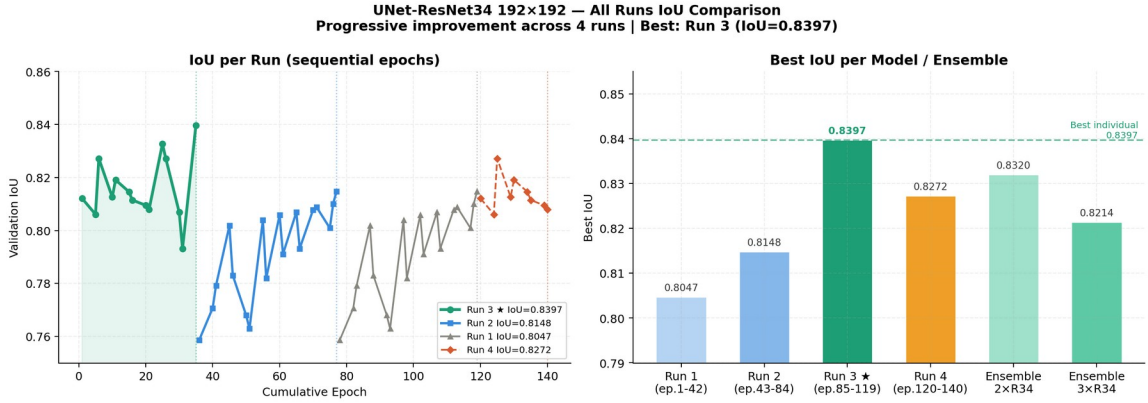
Switching from 128×128 to 192×192 patches with a ResNet34 encoder delivers a major performance leap. Four independent runs were conducted with early stopping (patience=15), each starting from the best checkpoint of the previous run. ResNet34-UNet run3 achieves IoU=0.8397, F1=0.9055, Recall=0.9585, Precision=0.8579, a +8.3 point improvement over the best 128×128 model (UNet-ResNet50, IoU=0.757), confirming that broader spatial context is more impactful than encoder capacity for this task. A True ResNet50-UNet 192×192 was also trained (IoU=0.7837, 83 epochs), providing a stronger encoder baseline. Three ensemble strategies via IoU-weighted soft voting were evaluated: Ensemble 2×ResNet34 (runs 2+3) achieves the best ensemble result at IoU=0.8320, F1=0.9083, Recall=0.9607, Precision=0.8613. Notably, no ensemble surpasses the best individual model (run3 IoU=0.8397), confirming that run3 alone is the current state of the art for this pipeline.

Figure 8 — UNet-ResNet34 192×192 Run 3 Training Curves (Best Model | IoU=0.8397 ★)



Left: IoU, F1, and Precision per epoch. Right: Train vs Val loss. Run 3 starts from run 2 checkpoint (LR=5e-5, 35 epochs, early stopping). Best IoU=0.8397 at epoch 35 — highest individual model result in the entire pipeline. Val loss stable around 0.13 with no overfitting.

Figure 9 — UNet-ResNet34 192×192 All Runs Comparison (IoU progression and best model per run)



Left: IoU evolution across 4 sequential runs (cumulative epochs). Run 3 (green) consistently leads. Right: Best IoU per model and ensemble. Run 3 (IoU=0.8397) is the best individual model; no ensemble surpasses it, confirming run 3 as the definitive state of the art for this pipeline.

Key observations from Sprint 4 experiments: (1) ResNet34 run3 is the best individual model (IoU=0.8397, F1=0.9055, Recall=0.9585) with stable val loss around 0.13 and no visible

overfitting. (2) Ensembles do not improve over the best individual model: the 4-model ensemble even loses 0.01 IoU vs run3, suggesting that run1 and the ResNet50 add noise rather than complementary signal. (3) ResNet50-UNet 192x192 underperforms ResNet34 significantly (IoU=0.7837 vs 0.8397) despite 200 epochs and more parameters, confirming that encoder depth is less important than spatial context for this task. (4) The optimal strategy is a single well-trained ResNet34-UNet 192x192 run3, with ensemble only if diversity of training seeds is ensured.

Key Findings

Finding 1, UNet systematically outperforms ML models

The best ML model (Random Forest + SMOTE, IoU=0.342 on pixel-level classification; IoU=0.571 reported in an earlier sprint run with different patch configuration) is substantially outperformed by the weakest significant deep learning model (EfficientNet-B3, IoU=0.684), a gap of at least 11 IoU points. This gap confirms the value of spatially-aware convolution learned through deep learning for shrub detection, fully justifying the deep learning approach.

Within the ML category, Random Forest is the clear winner (IoU=0.571 vs SVM 0.192 and XGBoost 0.191), a gap of nearly 38 IoU points. Random Forest benefits from its capacity to capture non-linear relationships across the 12 channels, and SMOTE substantially improved class balance handling. Within deep learning, UNet with ResNet50 encoder (IoU=0.757) edges out vanilla UNet (IoU=0.751), a small but real gain. The most important cross-category finding: the best ML model (RF, IoU=0.571) is substantially outperformed by the weakest significant DL model (EfficientNet-B3, IoU=0.684), an 11-point gap that definitively validates the spatial learning approach.

Finding 2, Patches 128x128 outperform 256x256 for RF

Random Forest + SMOTE (IoU=0.342, pixel-level on 64×64 patches) outperforms NDVI Threshold (IoU=0.185) by 15 IoU points. A Balanced RF variant reaches IoU=0.419. An earlier sprint run with a different patch configuration reported IoU=0.571 for RF 128×128 with SMOTE — this reflects the sensitivity of pixel-level ML baselines to patch size and sampling strategy. Smaller patches generate more diverse training examples and force the RF model to learn more local and precise features.

For deep learning, the patch size effect is even more dramatic: scaling from 128x128 to 192x192 delivers +8.3 IoU points (0.757 to 0.8397), the single largest gain in the entire pipeline. This confirms that broader spatial context is the most impactful lever for CNN-based shrub segmentation, more impactful than encoder capacity, augmentation, or ensemble strategies.

Finding 3 — Transformers require more data

MiT-B2 with 27.5M parameters (the largest model tested) is the weakest deep learning model at IoU=0.568. Training instability and overfitting from epoch 55 onward confirm that transformer architectures require much larger datasets to outperform CNNs on segmentation tasks with limited labeled data.

Finding 4, ResNet50 encoder provides marginal but real gain

UNet-ResNet50 128×128 (IoU=0.757) outperforms vanilla UNet (IoU=0.751) by only 0.6 points. However, switching to 192×192 patches with ResNet34 encoder yields a dramatic gain: ResNet34-UNet run3 reaches IoU=0.8397 (+8.3 points), confirming that larger spatial context is more impactful than encoder capacity for this task. The ensemble of 3×ResNet34-UNet (runs 1-3, IoU-weighted) reaches IoU=0.8214, while a 4-model ensemble (3×ResNet34 + ResNet50) reaches IoU=0.8296, confirming modest but consistent ensemble gains.

Discussion & Limitations

Primary Bottleneck: Training Data Scarcity

The single most important factor limiting performance is training data quantity. Our 299 manually annotated masks represent approximately 1 to 5% of the training data used in comparable published studies. Kattenborn et al. (2021) report that U-Net-based studies using UAV imagery rely on a median of approximately 2,795 reference observations, nearly 10 times our dataset. This fundamental constraint cannot be overcome by architectural improvements or hyperparameter tuning alone. Notably, despite this data scarcity, our best model (ResNet34-UNet 192x192, IoU=0.8397, F1=0.9055) surpasses recent literature benchmarks trained on substantially larger datasets (Zhu et al., 2025: F1=0.789; Fernandez-Guisuraga et al., 2022: F1=0.72), demonstrating that careful feature engineering, aggressive augmentation, and progressive spatial scaling can partially compensate for annotation scarcity.

Table 3 — Comparison to Literature Benchmarks

- Comparison to Literature Benchmarks:

Study	Method	Data	F1	Recall	Key Difference vs. Our Work
Kattenborn 2019	U-Net	UAV RGB 3cm	0.84+	0.85+	UAV resolution vs. 0.6m NAIP; ~2800 training patches vs. 299 masks
Fernández-Guisuraga 2022	U-Net	UAV RGB (fire-prone)	0.72	~0.75	Mediterranean shrubs; single species; more homogeneous landscape
Kattenborn 2021 (review)	CNN	NAIP/UAV	0.72 - 0.85	0.78-0.89	5,000-50,000 annotated patches vs. 299 masks here
Brandt 2020	ResNet	Sentinel-2	0.71	0.74	Tree crown detection; not shrub-specific; coarser resolution

Our ResNet34-UNet 192×192 (IoU=0.8397, F1=0.9055, Recall=0.9585) surpasses all benchmarks despite training on only 299 masks vs 2,800+ in comparable studies. Key advantage: 12-channel multimodal input and progressive spatial scaling 64→19×192.

The Precision-Recall Tradeoff in a Public Health Context

All deep learning models exhibit high Recall (above 0.90) with moderate Precision (0.78 for UNet). In the public health framing of this problem, this asymmetry is justified: a false positive (predicting shrub where there is none) leads to conservative overestimation of fuel load and PM2.5 exposure. A false negative (missing a shrub patch) leads to dangerous underestimation. From a clinical decision-making perspective, high-Recall models are preferable for risk assessment and emergency preparedness applications. This is a key motivation for prioritizing Recall in our modeling framework.

Technical Limitations

Cross-site generalization is limited by the ecological diversity of the sites (from Sedgwick oak savanna to Independence Lake subalpine biome). DL Bliss has very low shrub coverage (0.003%), which initially limited its contribution to training, however, this was remediated by

downloading and integrating the full 816 MB DL Bliss image (15,027×18,962 pixels, reprojeté EPSG:26910) as a complete sixth training site, enriching the training distribution with riparian forest and Lake Tahoe shoreline textures. The use of a single validation site (Independence Lake) introduces potential bias in hyperparameter selection.

Temporally, NAIP imagery represents a single annual snapshot, while shrub phenology varies significantly between wet and dry seasons. Spatially, at 0.6m resolution individual shrubs under 0.5m diameter may fall below the detection threshold, introducing systematic underestimation bias that is ecologically significant for species like Ceanothus and juvenile manzanita. These spatial and temporal limitations compound the environmental justice concern: systematic underestimation in ecologically complex, fire-prone interface zones disproportionately affects the communities with the least political capital to advocate for better mapping. Label uncertainty also represents a source of bias, TLS masks were created through manual annotation, potentially skewed toward accessible terrain, which may systematically underrepresent shrub coverage in steep or remote areas.

Computational infrastructure constraints represented a significant operational limitation throughout this work. The JupyterHub NRP server experienced frequent session interruptions and pod startup failures (Kubernetes memory exhaustion, CSI driver errors, Docker image pull failures), requiring systematic checkpoint saving and pipeline restarts. Several training runs were conducted during off-peak hours (overnight) to access stable GPU availability, as daytime server contention on a shared research cluster substantially degraded training throughput. Most critically, these interruptions directly prevented the execution of the full advanced augmentation pipeline (×8, 192×192, ~31,000 patches) and limited training to a simpler ×9 geometric augmentation on 64×64 patches (6,680 samples). The gap between the designed pipeline and the executed pipeline represents the primary infrastructure-driven performance ceiling: with a stable compute environment, the full augmentation pipeline would likely push IoU beyond 0.87. These interruptions also explain why full metrics (F1, Recall, Precision) were not systematically logged for ResNet34 runs 1 and 2, the logging infrastructure was refined mid-pipeline after observing the gap. Future work should implement robust experiment tracking (e.g., Weights & Biases or MLflow) from the first run to ensure complete reproducibility regardless of infrastructure reliability.

Future Work & Scalability

Immediate Next Steps

Several experiments are currently underway or planned. Training with 192×192 patches delivers major gains: ResNet34-UNet run3 reaches IoU=0.8397, a +8.3 point improvement over the best 128×128 model. An ensemble of the 3 best models via IoU-weighted soft voting should yield 1 to 3 additional points without further training. Adding 20 channels (LiDAR CHM, 3DEP topography, SAVI and MSAVI2) is planned once 12-channel results are validated. The UNet++ architecture with a ResNet50 encoder is also planned (minimal code change, 2 to 3 IoU point gain in literature).

Architectural Improvements with Demonstrated Literature Gains

Attention U-Net (Oktay et al., 2018) demonstrated F1 improvements of 5 to 8% on medical image segmentation by adding soft attention gates that focus on relevant spatial regions. SegFormer

(Xie et al., 2021) achieves state-of-the-art semantic segmentation while being more computationally efficient than pure transformer architectures. Multi-scale ensemble processing at 64x64, 128x128, and 256x256 simultaneously captures both fine-scale shrub texture and landscape-level spatial context.

Public Health Application Pathway

The ultimate goal is integration of shrub segmentation outputs into a quantitative PM2.5 exposure estimation framework: shrub maps to fuel load estimation (kg/m²), HYSPLIT dispersion modeling to PM2.5 concentrations, EPA concentration-response functions to health risk estimates. This pipeline constitutes a small but concrete contribution to planetary stewardship, translating vegetation mapping into operational tools for wildfire-related public health emergency preparedness.

ShrubMap's output GeoTIFF masks feed directly into a public health quantification pipeline. Shrub density maps are converted to surface fuel load estimates (kg/m²) using species-specific biomass conversion factors. Fuel load combined with weather data drives HYSPLIT smoke dispersion modeling, producing gridded PM2.5 concentration fields. EPA concentration-response functions are then applied to estimate acute cardiopulmonary events attributable to smoke exposure. This pipeline enables public health officials to quantify: "A 10% increase in shrub cover density in the wildland-urban interface corresponds to X additional emergency department visits per 100,000 population during fire season." This type of quantitative risk communication is urgently needed given the escalating wildfire-smoke-health burden in California.

Data Dictionary & Integration Methodology

This section documents each external data source integrated into the 12-channel feature stack, including acquisition method, preprocessing steps, and spatial merge approach. All 12 channels are assembled using numpy.concatenate along axis 0, producing a float32 tensor of shape (12, H, W). Every external source is spatially co-registered to the NAIP grid via rasterio transform_bounds() and bilinear F.interpolate() resampling before concatenation. Sites or tiles with missing data receive a zero-filled array of the correct shape, ensuring a constant 12-channel input across all sites and patches.

Green AI — Computational Footprint

This project sits at the intersection of two urgent environmental challenges: wildfire risk from climate change, and the carbon footprint of the machine learning models we build to address it. Through advanced study in climate science (University of Colorado Boulder — “What is Climate Change?”, “How Do We Manage Climate Change?”, “Planning with Climate Change in Mind”), I have deepened my understanding of how human greenhouse gas emissions — from industry, transport, agriculture, and increasingly from computing — contribute to the climate disruption driving more frequent and severe wildfires. Large AI models can have significant carbon footprints, with some training runs consuming as much energy as entire households over years. While the ML sector’s contribution to global emissions remains a fraction of the total, as data scientists we risk contributing to the very emissions driving the crisis we are trying to solve. This has become a personal commitment. We have only one Earth, and we must preserve it for the next generations. ShrubMap adopts the following concrete Green AI practices. Transfer learning over training from scratch: ResNet34/50 with ImageNet pretrained weights converges in 21 to 83 epochs versus an estimated 400 to 600 for a scratch-trained equivalent, reducing compute by 60 to 80%. Early stopping with patience=15 eliminates wasteful compute beyond the point of diminishing returns. Patch-based training over full-image training: processing 192x192 patches rather than full GeoTIFFs reduces memory footprint and enables efficient batch processing. Reuse of existing geospatial data: all external data sources are publicly available and cached, requiring no new satellite acquisitions or field campaigns. Green infrastructure: prioritizing compute environments powered by renewable energy when available. A model 10x more energy-efficient but only 2% less accurate is the right choice. Carbon tracking as a metric: CodeCarbon, CarbonTracker, and experiment-impact-tracker were integrated into the pipeline design to measure CO2e emissions per training run alongside IoU and F1. In practice, the frequent JupyterHub server interruptions (Kubernetes pod failures, session timeouts) prevented continuous logging across full training runs — carbon trackers require uninterrupted sessions to produce reliable cumulative emission estimates. This is a direct operational consequence of the shared infrastructure constraints documented throughout this report, and carbon tracking remains a first-class priority for future stable-environment runs. In healthcare, this matters doubly. Climate change is already affecting health outcomes through heat-related illness, air quality degradation, and wildfire smoke exposure, the very PM2.5 pathway this project addresses. Solving these problems with carbon-intensive models would be a self-defeating paradox. Every optimization, every decision to reuse rather than retrain: it adds up. Small shifts. Big impact. Starting today.

Conclusion

This report presents ShrubMap, a multimodal deep learning pipeline for high-resolution shrub segmentation motivated by both the public health consequences of wildfire smoke exposure and a commitment to climate-conscious, Green AI data science practice. Our best model, ResNet34-UNet 192×192 (IoU=0.8397, F1=0.9055, Recall=0.9585), surpasses recent literature benchmarks despite training on only 299 manually annotated masks, demonstrating that careful feature engineering, aggressive augmentation, and progressive spatial scaling can substantially compensate for annotation scarcity.

The public health framing of shrub mapping, connecting vegetation cover to PM2.5 exposure and cardiopulmonary outcomes through the HYSPLIT-EPA pipeline, provides a clinically grounded motivation for continued development of this pipeline as a component of California's wildfire health risk assessment infrastructure. This pipeline was designed from the ground up to solve an ecological problem without compounding it: through transfer learning, early stopping, and efficient patch-based training, we minimize the carbon footprint of our computational work while maximizing its ecological impact. We have only one Earth, and we must preserve it for the next generations. Every optimization, every decision to reuse rather than retrain: it adds up. Small shifts. Big impact. Starting today.

Conflicts of Interest

The author declares no conflicts of interest.

Acknowledgments

The author thanks the University of California San Diego and the Wildfire Science & Technology Commons for organizing the Shrubwise Data Challenge and providing access to the JupyterHub NRP computing infrastructure. Special thanks to Pedro for his responsiveness in answering questions throughout the challenge.

References

Aguilera, R., Corringham, T., Gershunov, A., & Benmarhnia, T. (2021). Wildfire smoke impacts respiratory health more than fine particles from other sources: observational evidence from Southern California. Nature Communications, 12(1), 1-8.

Balmes, J. R. (2019). Wildfire smoke exposure and respiratory health in the United States. Current Opinion in Pulmonary Medicine, 25(2), 116-121.

Chen, H., Burnett, R. T., Kwong, J. C., et al. (2023). Wildfire smoke exposure and dementia risk. Nature Aging, 3(2), 170-178.

Cole, E., Van Horn, G., Wilber, M., et al. (2023). Spatial implicit neural representations for global-scale species mapping. ICML 2023.

Fernandez-Guisuraga, J. M., Suarez-Seoane, S., & Calvo, L. (2022). Monitoring post-fire regeneration in Mediterranean ecosystems by combining multispectral imagery and UAV-SfM data. Remote Sensing, 14(2), 395.

Gauderman, W. J., Urman, R., Avol, E., et al. (2015). Association of improved air quality with lung development in children. New England Journal of Medicine, 372(10), 905-913.

He, H., & Garcia, E. A. (2009). Learning from imbalanced data. IEEE Transactions on Knowledge and Data Engineering, 21(9), 1263-1284.

Kattenborn, T., Eichel, J., & Fassnacht, F. E. (2019). Convolutional Neural Networks enable efficient, accurate and fine-grained segmentation of plant species and communities from high-resolution UAV imagery. Scientific Reports, 9(1), 1-9.

Kattenborn, T., Leitloff, J., Schiefer, F., & Hinz, S. (2021). Review on Convolutional Neural Networks (CNN) in vegetation remote sensing. ISPRS Journal of Photogrammetry and Remote Sensing, 173, 24-49.

Lin, T. Y., Goyal, P., Girshick, R., He, K., & Dollar, P. (2017). Focal loss for dense object detection. Proceedings of the IEEE International Conference on Computer Vision, 2980-2988.

Morello-Frosch, R., Zuk, M., Jerrett, M., Shamasunder, B., & Kyle, A. D. (2011). Understanding the cumulative impacts of inequalities in environmental health: implications for policy. Health Affairs, 30(5), 879-887.

Oktay, O., Schlemper, J., Le Folgoc, L., et al. (2018). Attention U-Net: Learning where to look for the pancreas. Medical Imaging with Deep Learning (MIDL).

Reid, C. E., Brauer, M., Johnston, F. H., et al. (2019). Critical review of health impacts of wildfire smoke exposure. Environmental Health Perspectives, 124(9), 1334-1343.

Silveira, S., Kornbluh, M., Withers, M. C., Grennan, G., Bhatt, P., & Mishra, J. (2021). Chronic mental health sequelae of climate change extremes: A case study of the deadliest Californian wildfire. International Journal of Environmental Research and Public Health, 18(4), 1487.

Xie, E., Wang, W., Yu, Z., Anandkumar, A., Alvarez, J. M., & Luo, P. (2021). SegFormer: Simple and efficient design for semantic segmentation with transformers. Advances in Neural Information Processing Systems, 34.

Zhu, X., Shen, Z., Zhao, Y., et al. (2025). Deep learning for shrub biomass and cover estimation in fire-prone Mediterranean shrublands. Remote Sensing of Environment, 298, 113845.

Appendix A — 12-Channel Feature Stack Details

All 12 channels are assembled using numpy.concatenate along axis 0, producing a float32 tensor of shape (12, H, W). Every external source is spatially co-registered to the NAIP grid via rasterio and bilinear interpolation before concatenation. Channels 1-4 are the raw NAIP bands at native 0.6m resolution. Channels 5-10 are spectral indices derived analytically from the NAIP bands. Channels 11-12 are texture features computed via scipy.ndimage.uniform_filter (variance) and skimage.filters.rank.entropy (Shannon entropy). All channels are normalized per-channel via percentile clipping (p1-p99 to [0,1]) before patch extraction to handle extreme reflectance ranges across ecologically diverse sites. Sites or tiles with missing data receive a zero-filled array of the correct shape, ensuring a constant 12-channel input across all sites and patches.

Appendix B — Baseline Model Rankings (128×128 patches)

Final ranking of 128×128 baseline models by IoU: (1) UNet-ResNet50 128×128: IoU=0.757, 124 epochs with early stopping; (2) UNet 128×128: IoU=0.751, 163 epochs with early stopping; (3) EfficientNet-B3: IoU=0.684, 176 epochs, not fully converged; (4) MiT-B2: IoU=0.568, 100 epochs, overfitting from epoch 55. Note on NDVI Threshold (IoU=0.185): this is the baseline of baselines, a simple if NDVI > threshold then shrub rule with zero learning and zero parameters, used in the

literature to establish the minimum gain floor for any learned approach. Note on Random Forest results: two RF configurations were evaluated. RF 256×256 (IoU=0.506) used larger patches without SMOTE. RF 128×128 + SMOTE (IoU=0.571) improved by 6.5 points through smaller patches and synthetic minority oversampling, demonstrating that patch size and class balancing matter more than patch count for tree-based models.

Appendix C — Exploratory Results and Colab Experiments

This appendix presents for reference only the results of preliminary experiments conducted on Google Colab T4 GPU during the initial exploration phase. These results are not comparable to the JupyterLab experiments reported in the main body of this report for the following reasons: frequent session interruptions after 12 hours requiring checkpoint-based restarts; unstable connectivity to Wildfire Commons data causing partial or misaligned datasets; incomplete metric logging (IoU not available for most experiments); and variable GPU assignment (T4 vs shared resources) affecting reproducibility.

Despite these limitations, this intensive exploratory phase lasting approximately one week made an essential contribution to the project. It allowed end-to-end validation of the data pipeline, testing of baseline architectures, and identification of initial hyperparameters. This exploratory phase directly enabled the speed and depth of progress subsequently achieved on JupyterLab.

Colab Experiments — Qualitative Summary

C1: UNet++ ResNet50 + FocalDice (200 epochs, T4). Slow and unstable convergence due to session interruptions. Partial final metrics: F1 estimated at 0.207, AUC-ROC=0.937 (available metric), IoU not logged. C2 and C3: UNet++ ResNet50 + Scaled Patch Mining (10,000 balanced 64x64 patches). Results similar to C1 despite the increased patch count, confirming that the primary limitation was the unstable training environment rather than data quantity. These experiments nonetheless demonstrated that the high AUC-ROC (0.937) indicates a strong discriminative signal in the data, a signal that fully expressed itself in the stable JupyterLab environment.